

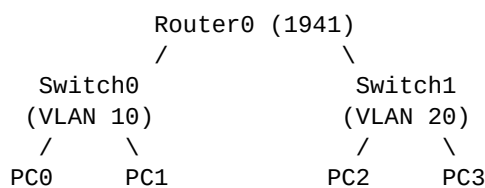
Inter-VLAN Routing Lab (Cisco Packet Tracer)

A Packet Tracer lab that demonstrates **inter-VLAN routing**: two departments on separate VLANs/subnets that can talk to each other only through a router.

- **File:** inter_routing_vlan.pkt
- **Tool:** Cisco Packet Tracer
- **Topic:** CCNA - VLANs and inter-VLAN routing

Topology

Device	Model	Role
Router0	Cisco 1941	Layer-3 gateway between the two VLANs
Switch0	Cisco 2960-24TT	Dept A switch (VLAN 10)
Switch1	Cisco 2960-24TT	Dept B switch (VLAN 20)
PC0, PC1	PC-PT	Dept A hosts
PC2, PC3	PC-PT	Dept B hosts



Addressing Plan

Host	VLAN	Department	IP Address
PC0	10	Dept A	192.168.10.11
PC1	10	Dept A	192.168.10.12
PC2	20	Dept B	192.168.20.13
PC3	20	Dept B	192.168.20.14

Subnet mask (assumed /24): 255 . 255 . 255 . 0

Default gateways (assumed): 192 . 168 . 10 . 1 for VLAN 10, 192 . 168 . 20 . 1 for VLAN 20.

The screenshot does not show masks, gateways or port numbers. Those values are the usual convention and should be checked against your .pkt file.

Sample Configuration

Switch0 (repeat on Switch1 with VLAN 20):

```
vlan 10
name DeptA
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
```

Router0 (one interface per switch):

```
interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
interface g0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
```

How to Test

- 1 Open inter routing vlan.pkt in Packet Tracer.
- 2 On each PC set IP, mask and default gateway (Desktop > IP Configuration).
- 3 From PC0 run ping 192.168.10.12 (same VLAN, should succeed).
- 4 From PC0 run ping 192.168.20.13 (other VLAN, succeeds only through the router).
- 5 Use tracert 192.168.20.14 to see the router hop.
- 6 Use Simulation mode to watch the ICMP packets cross the router.

Troubleshooting

- Ping fails across VLANs: check the router interfaces are no shutdown and have correct IPs.
- Wrong gateway on a PC: the PC cannot leave its own subnet.
- Port in the wrong VLAN: verify with show vlan brief.
- First ping times out: normal, it is ARP resolution; run it again.

Learning Outcomes

- Create VLANs and assign access ports.
- Explain why separate VLANs need a Layer-3 device to communicate.
- Configure router interfaces as default gateways.
- Verify connectivity with ping and tracert.